

JUNBAO LIANG

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SUMMARY

Computational chemistry Ph.D. candidate specializing in molecular dynamics. Research focuses on enzyme dynamics, electrostatic catalysis, and solvation. Experienced in polarizable and nonpolarizable simulations and force field parameterization. Four years of Linux and HPC experience. Develops automated Python workflows for simulation and data analysis.

TECHNICAL SKILLS

Molecular modeling: Polarizable/nonpolarizable MD; AMOEBA, CHARMM; Tinker9, GROMACS; Poltype2; MDAnalysis

Scientific computing: Python (NumPy, pandas, SciPy, Matplotlib); Bash; Linux; Slurm/HPC; Git

Trajectory analysis: RMSD, RMSF, RDF, SASA, hydrogen bond analysis, velocity autocorrelation function (VACF), dipole autocorrelation function (DACF), electric field analysis, block analysis, and cluster analysis

Quantum chemistry & visualization: Gaussian, VASP; VMD, PyMOL

EDUCATION

Virginia Tech — Ph.D. in Chemistry, GPA 3.85 | Expected May 2027

M.S. Chemistry, University of Oregon (2020) | **B.S. Applied Chemistry**, CUMT (2019)

RESEARCH EXPERIENCE | Graduate Research Assistant, Virginia Tech | Aug 2022–Present

Electrostatic Preorganization and Mg²⁺ Positioning in β -Phosphoglucomutase | First author manuscript in major revision

- Led 12 AMOEBA simulations of wild type and Asp10Asn mutant β phosphoglucomutase in ground state and transition state/product state configurations. Each polarizable MD simulation contained approximately 43,000 atoms and ran for 250 ns, totaling 3.0 μ s.
- Generated conformers with Rosetta, parameterized ligands with Poltype2, and decomposed projected electric fields along four reaction coordinates with Python and ELECTRIC.
- Developed GROMOS cluster analysis using structural RMSD, total projected electric fields, and electric field contributions from individual residues. Used block analysis of active site distances and angles to verify convergence and cluster reliability.
- Cluster centroids revealed Mg²⁺ mispositioning in the Asp10Asn mutant and showed disruption of electrostatic preorganization in the ground state beyond the previously proposed loss of proton donation.

Molecular Mechanism of DszB Glycoengineering

- Designed, automated, and completed 27 polarizable AMOEBA simulations in Tinker9 using Python, Bash, and Slurm.
- The study covered wild type DszB and DszB glycosylated at serine or threonine in apo, reactant bound, and product bound states. Each system contained more than 70,000 atoms, totaling 4.05 μ s.
- Generated AMOEBA parameters for the glycosylated residues and ligands with Poltype2 using Psi4 for MP2 calculations. Developed Python and MDAnalysis workflows for structural changes, loop flexibility, solvent exposure, interaction energies, radial distribution, and convergence.
- The analyses identified serine as the effective glycosylation site. Glycosylation at this site strengthened interactions between the loop and water, increased loop flexibility, promoted product release, and preserved reactant positioning.

Systematic Benchmark of Peptide Hydration Dynamics | Manuscript in preparation

- Independently designed, built, and automated a dataset containing 120 polarizable AMOEBA simulations in Tinker9. It covers all 20 amino acid homopentamers and 10 heterogeneous pentamers at 0, 0.5, 1, and 2 M glucose.
- The heterogeneous set includes five pairs with identical compositions but different sequence orders. Each system contained approximately 20,000 atoms and ran for 100 to 120 ns, totaling 12.0 to 14.4 μ s.
- Developed SciPy workflows to fit dipole autocorrelation functions with a biexponential model and integrate velocity autocorrelation functions to calculate diffusion coefficients. Combined these measurements with hydrogen bond network and electric field analyses.
- Found that hydration water became progressively slower as glucose concentration increased, consistent with a stronger dehydration effect. Showed that sequence order changes intramolecular hydrogen bonding, peptide conformation, and surrounding water dynamics.

Molecular Dynamics of Sodium Ion Battery Electrolytes | Second author manuscript in preparation

- Led the molecular dynamics component of an experimental collaboration. Compared polarizable AMOEBA and nonpolarizable CHARMM simulations of 1.0 M sodium triflate, 1.0 M sodium hexafluorophosphate, and 0.6 M sodium hexafluorophosphate in diglyme.
- Completed six simulations of approximately 45,000 atoms for 100 ns each.
- Calculated diffusion coefficients from mean square displacement. Used radial distribution functions and coordination numbers to characterize ion solvation.
- Identified 0.6 M sodium hexafluorophosphate as the fastest diffusing formulation, with a sodium ion diffusion coefficient of 1.33×10^{-6} cm²/s. The simulation results agreed with infrared spectroscopy, nuclear magnetic resonance, and electrochemical measurements.

SELECTED PUBLICATION | Liang, J., Zheng, Y., & Vaissier Welborn, V. "Mitigating product inhibition in 2'-hydroxybiphenyl-2-sulfinate desulfinase (DszB) with synthetic glycosylation." Protein Science 34(7), e70187 (2025). [doi:10.1002/pro.70187](https://doi.org/10.1002/pro.70187)